

```
In [1]: import pandas as pd
import numpy as np
df = pd.read_csv("train.csv")

print("Dataset Shape:", df.shape)
df.head()
```

Dataset Shape: (891, 12)

Out[1]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



```
In [2]: print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
10   Cabin        204 non-null    object
11   Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

```

Missing Values:

```

PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

```

In [3]: duplicates = df.duplicated().sum()

print("Duplicate Rows:", duplicates)
df = df.drop_duplicates()

print("Shape After Removing Duplicates:", df.shape)

```

```

Duplicate Rows: 0
Shape After Removing Duplicates: (891, 12)

```

```

In [4]: df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df.drop("Cabin", axis=1, inplace=True)
print(df.isnull().sum())

```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

```
In [5]: df["Survived"] = df["Survived"].astype("int")
df["Pclass"] = df["Pclass"].astype("int")
print(df.dtypes)
```

```
PassengerId    int64
Survived       int64
Pclass         int64
Name           object
Sex            object
Age            float64
SibSp          int64
Parch          int64
Ticket         object
Fare           float64
Embarked       object
dtype: object
```

```
In [6]: df.rename(columns={
    "PassengerId": "Passenger_ID",
    "Pclass": "Passenger_Class",
    "SibSp": "Siblings_Spouses",
    "Parch": "Parents_Children"
}, inplace=True)
print(df.columns)
```

```
Index(['Passenger_ID', 'Survived', 'Passenger_Class', 'Name', 'Sex', 'Age',
      'Siblings_Spouses', 'Parents_Children', 'Ticket', 'Fare', 'Embarked'],
      dtype='object')
```

```
In [7]: print(df.info())

df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Passenger_ID          891 non-null   int64
1   Survived              891 non-null   int64
2   Passenger_Class       891 non-null   int64
3   Name                  891 non-null   object
4   Sex                   891 non-null   object
5   Age                   891 non-null   float64
6   Siblings_Spouses     891 non-null   int64
7   Parents_Children     891 non-null   int64
8   Ticket                891 non-null   object
9   Fare                  891 non-null   float64
10  Embarked              891 non-null   object
dtypes: float64(2), int64(5), object(4)
memory usage: 76.7+ KB
None
```

Out[7]:

	Passenger_ID	Survived	Passenger_Class	Name	Sex	Age	Siblings_Spouses	Parents_Children
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0

```
In [8]: df.to_csv("cleaned_titanic.csv", index=False)

print("Cleaned dataset saved successfully.")

Cleaned dataset saved successfully.
```